

HOPEN, Svalbard and Jan MayenWMO: **010620**Lat: **76.509N**Lon: **25.013E**Elev: **10**StdP: **101.20**Time Zone: **1.00 (EUC)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
3	-27.1	-24.9	-29.4	0.2	-26.9	-27.0	0.3	-24.8	15.1	-9.0	13.6	-7.7	5.7	40	1.006

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	2.2	7.3	6.5	6.6	5.8	5.9	5.3	6.7	7.1	6.0	6.4	5.4	5.8	6.5	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6.3	5.9	7.0	5.6	5.6	6.3	5.0	5.4	5.7	22.2	7.2	20.7	6.5	19.5	5.9	11.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 13.4	12.0	10.8	DB	-25.3	9.8	5.4	2.1	-29.2	11.3	-32.4	12.6	-35.5	13.7	-39.4	15.3	(4)
(5)			WB	-25.6	8.5	5.3	1.4	-29.4	9.5	-32.5	10.3	-35.5	11.1	-39.3	12.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	-3.4	-8.3	-9.0	-10.3	-8.1	-3.1	0.7	3.2	3.7	2.4	-1.0	-4.3	-6.9	
	DBStd	7.00	7.74	7.45	6.97	5.77	3.44	1.74	1.70	1.56	2.05	3.40	5.06	6.67	
	HDD10.0	4883	568	531	628	542	407	279	211	194	228	343	429	523	
	HDD18.3	7925	826	764	886	792	665	529	469	452	478	601	679	782	
	CDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	
	CDH23.3	0	0	0	0	0	0	0	0	0	0	0	0	0	
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0	
Wind	WSAvg	5.5	6.2	6.2	5.8	5.3	4.9	4.4	5.1	4.8	5.3	5.9	6.2	6.1	
Precipitation	PrecAvg	431	36	37	36	28	21	27	34	40	46	45	42	39	
	PrecMax	753	79	135	73	64	71	65	102	99	77	90	98	100	
	PrecMin	314	11	4	7	8	6	4	4	12	13	20	12	14	
	PrecStd	92	18	24	18	14	14	16	21	21	16	18	21	20	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	2.7	2.6	2.1	2.6	3.4	6.1	8.5	8.8	7.7	6.2	4.0	3.1	
		MCWB	1.9	1.8	1.4	1.7	2.5	5.0	7.5	7.9	6.8	5.4	3.0	2.2	
	2%	DB	1.7	1.5	1.0	1.5	2.2	4.8	7.1	7.4	6.6	5.2	2.7	2.1	
		MCWB	1.1	0.8	0.4	0.9	1.4	3.7	6.3	6.6	6.0	4.6	2.0	1.4	
	5%	DB	0.5	0.2	-0.1	0.6	1.4	3.9	6.3	6.6	5.9	4.1	1.7	1.0	
		MCWB	-0.1	-0.5	-0.5	0.1	0.7	3.0	5.5	5.9	5.4	3.5	1.1	0.3	
	10%	DB	-0.5	-0.7	-0.9	-0.3	0.8	3.1	5.6	5.9	5.1	3.1	0.9	0.0	
		MCWB	-1.1	-1.2	-1.2	-0.8	0.1	2.3	4.9	5.3	4.6	2.5	0.4	-0.5	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.2	2.1	1.6	2.0	2.7	5.2	7.7	8.2	7.1	5.7	3.3	2.5
			MCDB	2.6	2.6	2.0	2.4	3.1	5.9	8.3	8.7	7.5	6.0	3.8	3.0
2%		WB	1.2	1.0	0.5	1.0	1.7	4.1	6.5	6.9	6.2	4.7	2.2	1.5	
		MCDB	1.7	1.5	0.9	1.4	2.2	4.7	6.9	7.3	6.6	5.3	2.6	2.0	
5%		WB	0.0	-0.4	-0.5	0.2	0.9	3.2	5.7	6.1	5.5	3.6	1.3	0.5	
		MCDB	0.4	0.0	-0.1	0.6	1.3	3.8	6.2	6.5	5.9	4.1	1.7	0.9	
10%		WB	-1.0	-1.2	-1.2	-0.7	0.3	2.5	5.1	5.4	4.7	2.6	0.5	-0.4	
		MCDB	-0.6	-0.8	-0.9	-0.3	0.7	3.1	5.5	5.8	5.1	3.1	0.9	0.0	
Mean Daily Temperature Range	5% DB	MDBR	4.3	4.6	4.7	4.4	3.0	2.3	2.5	2.2	2.0	2.5	3.3	4.0	
		MCDBR	4.0	4.1	4.6	4.0	3.2	3.3	3.5	3.2	3.0	3.1	3.4	4.1	
		MCWBR	3.9	4.1	4.7	4.0	2.9	2.8	3.0	2.8	2.9	3.3	3.5	4.1	
	5% WB	MCDBR	3.9	4.2	4.7	4.0	3.1	3.1	3.3	3.1	2.9	3.1	3.5	4.1	
		MCWBR	4.0	4.2	4.8	4.0	3.0	2.7	3.0	2.8	2.8	3.3	3.5	4.1	
Clear-Sky Solar Irradiance	taub	N/A	0.226	0.275	0.304	0.321	0.307	0.318	0.311	0.284	0.356	N/A	N/A		
	taud	N/A	1.899	2.060	2.170	2.286	2.455	2.547	2.544	2.454	2.646	N/A	N/A		
	Ebn at Noon	0	89	564	755	816	856	831	775	621	132	0	0		
	Edh at Noon	0	12	59	88	94	84	74	63	43	13	0	0		
All-Sky Solar Radiation	RadAvg	0.00	0.05	0.82	2.64	3.87	4.85	3.73	2.35	0.90	0.13	0.00	0.00		
	RadStd	0.00	0.01	0.15	0.47	0.59	0.70	0.38	0.26	0.09	0.02	0.00	0.00		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days					
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only		+1.56	+4.63	+4.00	N/A	N/A	N/A	-539	-538	N/A	N/A		
(47) Regional (0 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Nomenclature: See separate page